WORKED SOLUTION

STEP I 1988, Question 3

Pure • Geometric Extremal Bounds

STEP I 1988, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s1-q3

The key idea

Ask where a point of the square can be if it is *far* from O: it must be squeezed into a small region near the opposite corner. If both P and Q are far from O, they are both in that small region, so they are close to each other. The number $\sqrt{6}-\sqrt{2}$ is exactly the size that makes this work, and it comes from an equilateral triangle sitting symmetrically in the square.

Solution

Work in centimetres with O at the origin and the square $0 \le x \le 1$, $0 \le y \le 1$. Write

$$s = \sqrt{6}-\sqrt{2}, \qquad s^2 = 6+2-2\sqrt{12} = 8-4\sqrt{3} \approx 1.072,$$

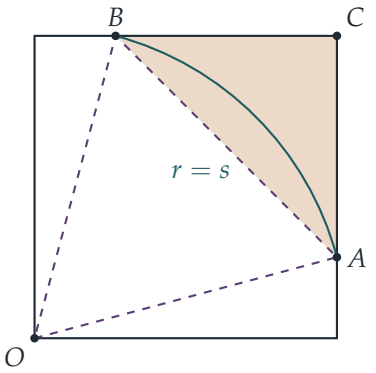
and let $t = 2-\sqrt{3} \approx 0.268$. Define the points

$$A = (1,t), \qquad B = (t,1), \qquad C = (1,1).$$

Then

$$\begin{aligned} OA^2 = OB^2 &= 1+(2-\sqrt{3})^2 = 8-4\sqrt{3} = s^2, \\ AB^2 &= 2(1-t)^2 = 2(\sqrt{3}-1)^2 = 8-4\sqrt{3} = s^2, \\ AC = BC &= 1-t = \sqrt{3}-1 \approx 0.732 < s. \end{aligned}$$

So OAB is an equilateral triangle of side s inside the square.



The teal arc is the circle of radius s about O; the part of the square beyond it lies inside the shaded triangle ABC.

Step 1: points far from O lie in triangle ABC

The line AB is $x + y = 1 + t$. Consider the part of the square on the O side of this line, $\{x + y \leq 1 + t\}$. It is the pentagon with vertices

$$O = (0,0), \quad (1,0), \quad A, \quad B, \quad (0,1).$$

The function $x^2 + y^2$ is convex, so its maximum over this convex pentagon is attained at a vertex. The values at the vertices are 0, 1, s^2, s^2, 1, and the largest is s^2. Hence

$$x + y \leq 1 + t \implies x^2 + y^2 \leq s^2$$

for every point of the square. Turning this round: if a point (x, y) of the square has distance from O *greater* than s, then $x + y > 1 + t$, and since $x \leq 1$ and $y \leq 1$ the point lies in the triangle ABC.

> **Ask yourself** Why is the largest distance between two points of a triangle one of its sides?

Fix Q. The distance $|PQ|$ is a convex function of P, so over the triangle it is largest when P is a vertex. Now fix that vertex and vary Q: by the same argument Q may also be taken to be a vertex. So the largest distance between two points of a triangle is the distance between two vertices, which is its longest side.

Step 2: the contradiction

Suppose that $OP > s$ and $OQ > s$; otherwise we are already done. By Step 1, both P and Q lie in triangle ABC. The longest side of that triangle is $AB = s$ (the other two sides are $\sqrt{3} - 1 < s$). So by the box above,

$$PQ \leq s.$$

In every case, then, at least one of OP, PQ, QO is at most $s = (\sqrt{6} - \sqrt{2})$ cm, which is what we had to show.

The bound is best possible

Taking $P = A$ and $Q = B$ gives $OP = PQ = QO = s$ exactly, so no smaller number would do. Since $\cos 15^\circ = \frac{\sqrt{6}+\sqrt{2}}{4}$, we have $s = \sec 15^\circ$: the extreme triangle is the equilateral triangle placed symmetrically about the diagonal, with each of OA and OB making 15° with a side of the square.

Checks

- $s \approx 1.0353$, a little more than the side of the square and well under the diagonal $\sqrt{2} \approx 1.414$, which is sensible.
- Numerically, 10^6 random pairs P, Q in the square never gave $\min(OP, PQ, QO)$ above s (the largest was about 1.0235). A hill-climbing search for the largest possible minimum settled on $P \approx (1, 0.26795)$, $Q \approx (0.26795, 1)$ with value 1.035276, matching A, B and s.

Common slips

- Arguing only that $\angle POQ < 60^\circ$ when $OP, OQ > s$. That is true, but it does not by itself make PQ the shortest side, so it does not finish the proof without more work.
- Checking only the symmetric configuration. It shows the bound cannot be improved; it does not prove that the bound holds for every P and Q.

Beyond STEP

This is a small extremal problem about packing points: how far apart can three points of a region be, all at once? Questions like this lead to packing and covering problems, and to the idea of the *diameter* of a set, which is the same for a set as for its convex hull. That fact is what made Step 2 painless. For more problems in this spirit, see Siklos, *Advanced Problems in Mathematics*.

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